

WILKERSON PASS, CO, USA

WMO: 726396

Lat: 39.050N

Lon: 105.516W

Elev: 11280

StdP: 9.61

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 00422

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	1.3	-32.6	1.3	28.8	-23.8	2.3	30.3	24.6	24.3	22.5	22.5	10.1	320	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	72.6	45.3	70.1	45.1	67.5	44.4	51.4	61.3	50.0	60.5	48.5	59.2	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.9	75.7	54.7	45.9	70.3	52.2	45.0	67.9	51.5	25.6	61.9	24.6	60.3	23.6	58.7	57.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 18.8	(c) 16.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 20.9	(5) 18.8	(6) 16.9	(7) -8.1	(8) 54.1	(9) 3.5	(10) 1.7	(11) -10.6	(12) 55.3	(13) -12.6	(14) 56.3	(15) -14.6	(16) 57.2	(17) -17.1	(18) 58.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	21.2	22.4	28.8	34.0	41.3	53.7	58.1	55.2	50.0	40.4	31.0	22.5		
	DBStd	15.02	8.18	9.07	8.20	7.51	7.72	6.50	4.90	4.82	6.21	8.26	9.07	8.76		
	HDD50	4938	893	773	656	481	284	38	5	12	69	303	570	854		
	HDD65	9752	1358	1193	1121	931	734	341	219	305	449	762	1020	1319		
	CDD50	669	0	0	0	0	14	149	255	174	70	7	0	0		
	CDD65	7	0	0	0	0	0	1	4	2	0	0	0	0		
	CDH74	44	0	0	0	0	0	10	23	11	0	0	0	0		
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0		
	WSAvg	8.2	9.7	10.3	8.9	8.6	7.1	6.9	5.9	5.9	7.2	8.3	9.8	10.1		
	PrecAvg	18.3	0.8	0.9	1.6	2.2	2.1	1.6	2.4	2.4	1.6	1.1	0.9	0.8		
(15) Precipitation	PrecMax	24.6	1.9	2.2	4.7	5.0	6.0	3.1	4.5	4.4	7.7	2.6	2.4	2.4		
	PrecMin	12.0	0.3	0.1	0.0	0.3	0.8	0.2	0.5	1.0	0.0	0.2	0.3	0.2		
	PrecStd	3.0	0.4	0.5	1.0	1.2	1.1	0.9	1.1	0.8	1.3	0.6	0.5	0.5		
	PrecStd	3.0	0.4	0.5	1.0	1.2	1.1	0.9	1.1	0.8	1.3	0.6	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	45.3	53.4	57.3	65.8	75.1	77.0	74.9	69.7	63.1	54.1	42.9		
		MCWB	27.7	29.4	32.8	32.2	39.4	42.7	45.8	47.3	42.7	39.1	32.0	27.4		
	2%	DB	36.7	39.4	47.8	53.7	61.2	71.9	73.0	70.4	65.6	57.4	48.4	37.1		
		MCWB	25.4	26.8	31.0	32.7	38.7	43.0	45.7	47.0	42.7	36.9	31.4	25.6		
	5%	DB	34.1	36.3	44.6	48.4	56.6	68.4	70.5	66.4	63.3	54.5	45.4	34.4		
		MCWB	24.4	25.6	29.5	32.4	37.4	42.5	46.3	46.1	42.0	36.6	30.0	24.4		
	10%	DB	31.8	33.6	40.6	45.2	53.7	64.3	67.7	63.7	60.8	51.8	41.3	32.2		
		MCWB	23.5	24.5	27.6	30.7	37.1	42.1	45.8	45.6	41.5	35.6	28.0	23.3		
	0.4%	WB	29.5	31.9	34.3	37.4	43.5	50.0	54.0	54.4	49.5	44.0	34.3	30.0		
		MCDB	36.9	41.9	46.8	50.0	56.4	61.6	63.1	64.8	60.4	54.8	46.4	37.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.1	28.5	32.2	34.9	41.3	47.7	51.6	51.5	47.2	41.7	32.4	27.8		
		MCDB	34.4	36.9	45.0	45.4	54.0	59.6	61.8	60.7	57.1	51.9	44.4	34.2		
	5%	WB	25.3	26.8	30.4	33.2	39.5	46.1	50.3	50.1	45.4	39.3	30.9	26.1		
		MCDB	32.6	34.4	41.5	43.7	51.7	58.3	61.4	59.5	55.3	48.5	42.5	32.2		
	10%	WB	23.7	24.9	28.6	31.9	38.3	44.7	48.9	48.6	43.7	37.2	29.3	24.4		
		MCDB	30.5	32.0	38.1	42.3	50.1	57.5	60.4	58.1	54.9	46.6	40.1	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	15.9	17.7	18.6	18.4	21.2	19.2	18.9	19.6	17.5	15.5	13.4		
		MCDBR	13.9	16.7	19.2	22.0	22.5	24.2	21.8	21.9	21.5	20.1	17.0	13.3		
		MCWBR	8.8	10.1	9.9	10.1	9.0	8.7	7.3	7.8	8.5	9.1	8.6	8.0		
		MCDBR	13.9	15.2	18.0	18.6	20.5	21.5	19.1	18.7	18.8	18.4	16.0	13.2		
	5% WB	MCWBR	9.6	10.0	9.9	9.4	9.2	9.1	8.4	8.3	8.5	8.9	8.7	9.0		
		MCWBR	9.6	10.0	9.9	9.4	9.2	9.1	8.4	8.3	8.5	8.9	8.7	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.193	0.197	0.216	0.228	0.243	0.254	0.275	0.269	0.242	0.221	0.202	0.194		
	taud		2.697	2.703	2.662	2.671	2.693	2.666	2.641	2.638	2.732	2.750	2.732	2.691		
	Ebn at Noon		331	340	338	336	329	324	316	317	324	325	325	324		
	Edh at Noon		19	22	26	28	28	29	29	28	24	21	18	18		
(44) All-Sky Solar Radiation	RadAvg		897	1240	1602	1839	1988	2229	1994	1821	1656	1273	931	769		
	RadStd		36	63	126	122	155	122	101	90	99	77	52	41		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+34

Nomenclature: See separate page